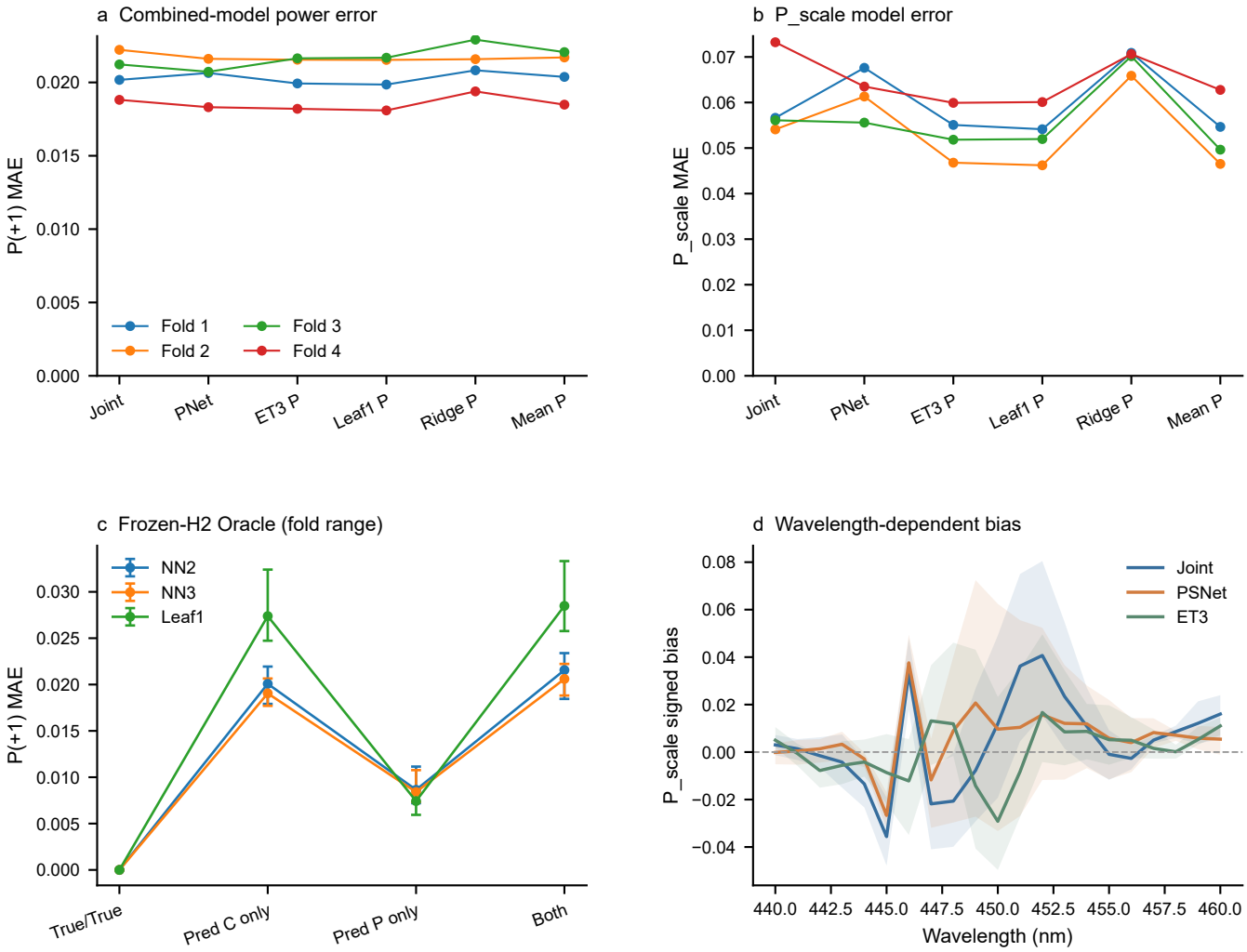
a Combined-model power error
P(+1) MAE
0.020
0.015
0.010
0.005
0.000
Fold 1
Fold 2
Fold 3
Fold 4
Joint
PNet
ET3 P
Leaf1 P
Ridge P
Mean P
b P_scale model error
P_scale MAE
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
Joint
PNet
ET3 P
Leaf1 P
Ridge P
Mean P
c Frozen-H2 Oracle (fold range)
P(+1) MAE
NN2
NN3
Leaf1
0.030
0.025
0.020
0.015
0.010
0.005
0.000
True/True
Pred C only
Pred P only
Both
d Wavelength-dependent bias
P_scale signed bias
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
Joint
PSNet
ET3
440.0
442.5
445.0
447.5
450.0
452.5
455.0
457.5
460.0
Wavelength (nm)